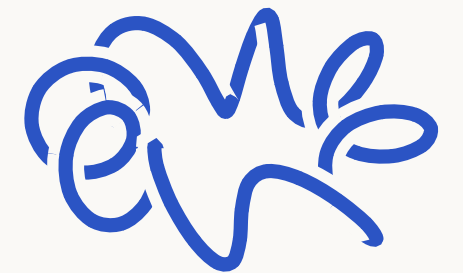


The Stable Andrews–Curtis Conjecture



change of variables and heuristic search for evaluations

Avigya Paudel

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Lucas Fagan · Giorgi Butbaia · Michele Tarquini

Both are balanced presentations of the trivial group

$$X = x^{-1} \cdot Y = y^{-1}$$

$$\langle x, y \mid x, y \rangle$$

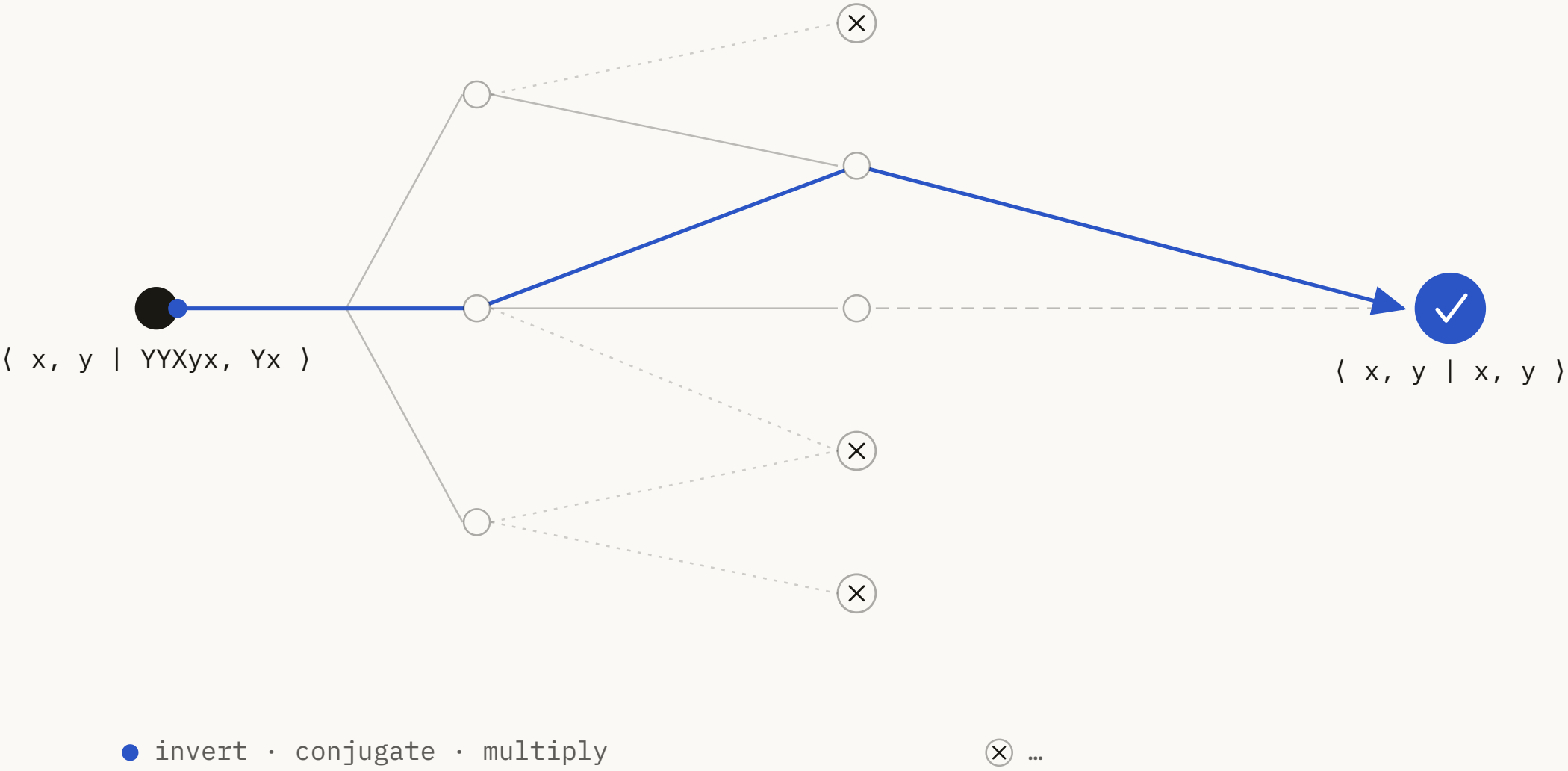
the trivial presentation · obviously trivial

$$\langle x, y \mid YYxyx, Yx \rangle$$

the trivial group · provably

Does a path always exist?

three moves · 1965



Three moves

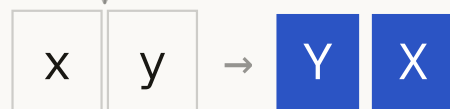
elementary AC · repeat

1

invert

$$r_i \rightarrow r_i^{-1}$$

$\langle x, y \mid xy, Yx \rangle$



$\langle x, y \mid YX, Yx \rangle$

2

conjugate

$$r_i \rightarrow g r_i g^{-1}$$

$$g \in \{x, y, X, Y\}$$

$\langle x, y \mid xy, Yx \rangle$



$\langle x, y \mid xxyX, Yx \rangle$

3

multiply

$$r_i \rightarrow r_i r_j$$

$\langle x, y \mid xy, Yx \rangle$



$\langle x, y \mid xx, Yx \rangle$

An example

three moves · total length 5

start

$\langle x, y \mid$ y X $,$ X y x $\rangle$



An example

three moves · total length 5

start

$$\langle x, y \mid \boxed{y} \boxed{X}, \boxed{X} \boxed{y} \boxed{x} \rangle$$

1 invert
 $\mathbb{I}_1 \rightarrow \mathbb{I}_1^{-1}$

$$\langle x, y \mid \boxed{x} \boxed{Y}, \boxed{X} \boxed{y} \boxed{x} \rangle$$



An example

three moves · total length 5

start

$\langle x, y \mid \boxed{y} \boxed{X}, \boxed{X} \boxed{y} \boxed{x} \rangle$

① invert
 $r_1 \rightarrow r_1^{-1}$

$\langle x, y \mid \boxed{x} \boxed{Y}, \boxed{X} \boxed{y} \boxed{x} \rangle$

② conjugate
 $r_2 \rightarrow x r_2 x^{-1}$

$\langle x, y \mid \boxed{x} \boxed{Y}, \boxed{x} \boxed{X} \boxed{y} \boxed{x} \boxed{X} \rangle$

rotation

rot_1

$\boxed{x} \boxed{y} \boxed{x} \rightarrow \boxed{y} \boxed{x} \boxed{x}$



An example

three moves · total length 5

start

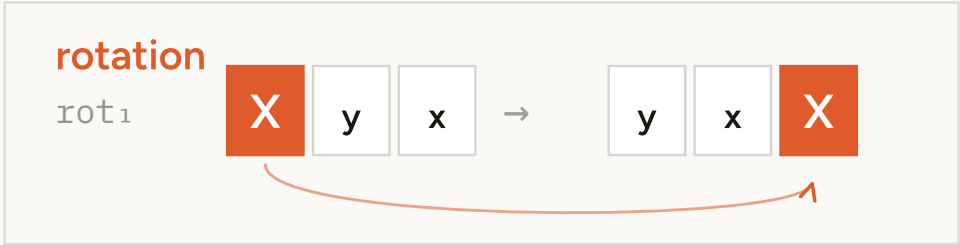
$$\langle x, y \mid \boxed{y} \boxed{X}, \boxed{X} \boxed{y} \boxed{x} \rangle$$

1 invert
 $r_1 \rightarrow r_1^{-1}$

$$\langle x, y \mid \boxed{x} \boxed{Y}, \boxed{X} \boxed{y} \boxed{x} \rangle$$

2 conjugate
 $r_2 \rightarrow x r_2 x^{-1}$

$$\langle x, y \mid \boxed{x} \boxed{Y}, \boxed{y} \rangle$$



An example

three moves · total length 5

start

$$\langle x, y \mid \boxed{y} \boxed{X}, \boxed{X} \boxed{y} \boxed{x} \rangle$$

1 invert
 $r_1 \rightarrow r_1^{-1}$

$$\langle x, y \mid \boxed{x} \boxed{Y}, \boxed{X} \boxed{y} \boxed{x} \rangle$$

2 conjugate
 $r_2 \rightarrow x r_2 x^{-1}$

$$\langle x, y \mid \boxed{x} \boxed{Y}, \boxed{y} \rangle$$

3 multiply
 $r_1 \rightarrow r_1 r_2$

$$\langle x, y \mid \boxed{x} \boxed{Y} \boxed{y}, \boxed{y} \rangle$$



An example

three moves · total length 5

start

$$\langle x, y \mid \boxed{y} \boxed{X}, \boxed{X} \boxed{y} \boxed{x} \rangle$$

1 invert
 $r_1 \rightarrow r_1^{-1}$

$$\langle x, y \mid \boxed{x} \boxed{Y}, \boxed{X} \boxed{y} \boxed{x} \rangle$$

2 conjugate
 $r_2 \rightarrow x r_2 x^{-1}$

$$\langle x, y \mid \boxed{x} \boxed{Y}, \boxed{y} \rangle$$

3 multiply
 $r_1 \rightarrow r_1 r_2$

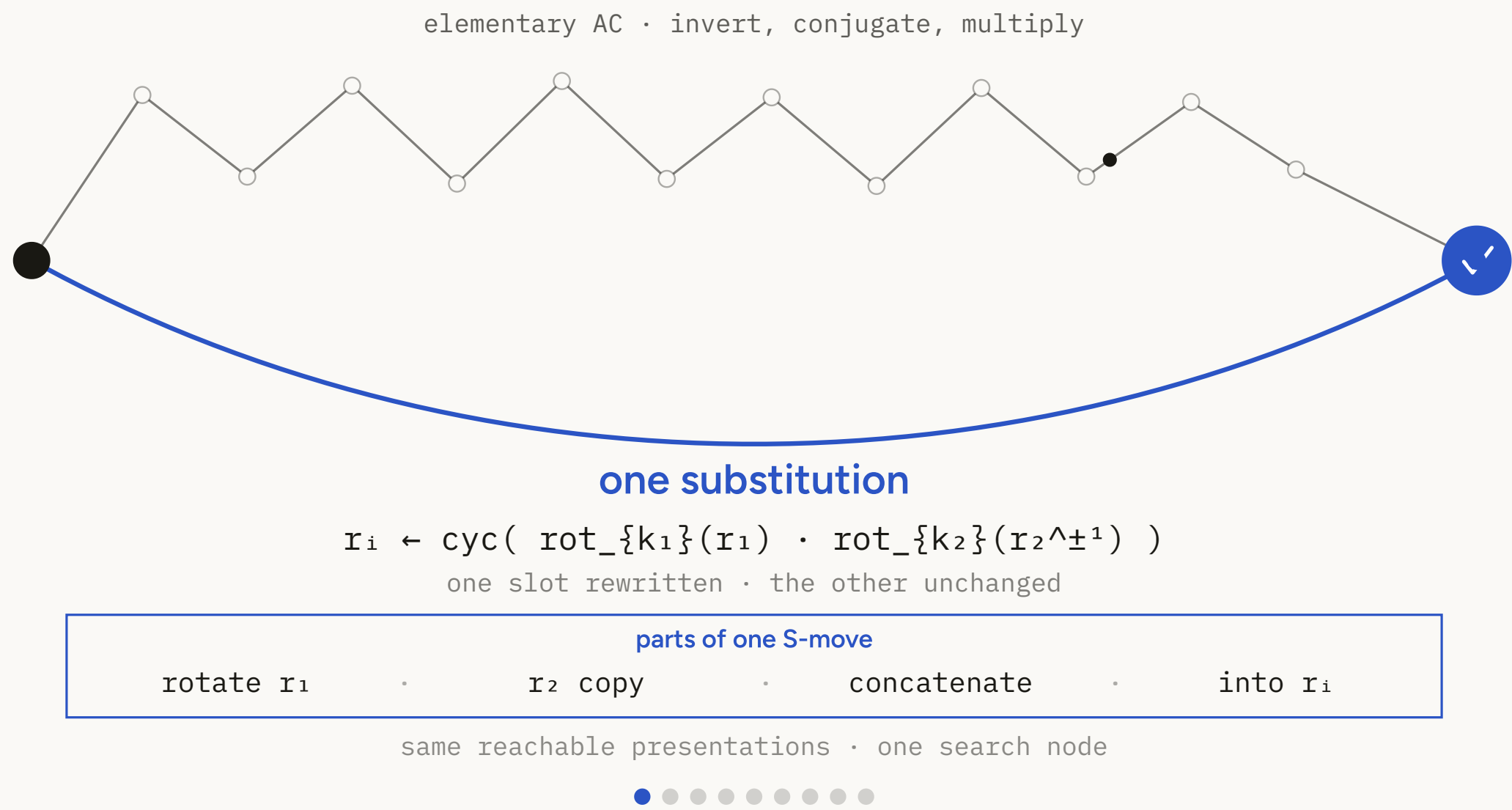
$$\langle x, y \mid \boxed{x}, \boxed{y} \rangle$$

trivial



One S-move

then to trivial



One S-move

then to trivial



$\{ YYXyx, YYx \}$

$$r_1 \leftarrow \text{rot}_3(r_1) \cdot \text{rot}_2(r_2 \text{ copy})$$

one S-move

$$k_1 = 3 \cdot k_2 = 2$$

into r_1

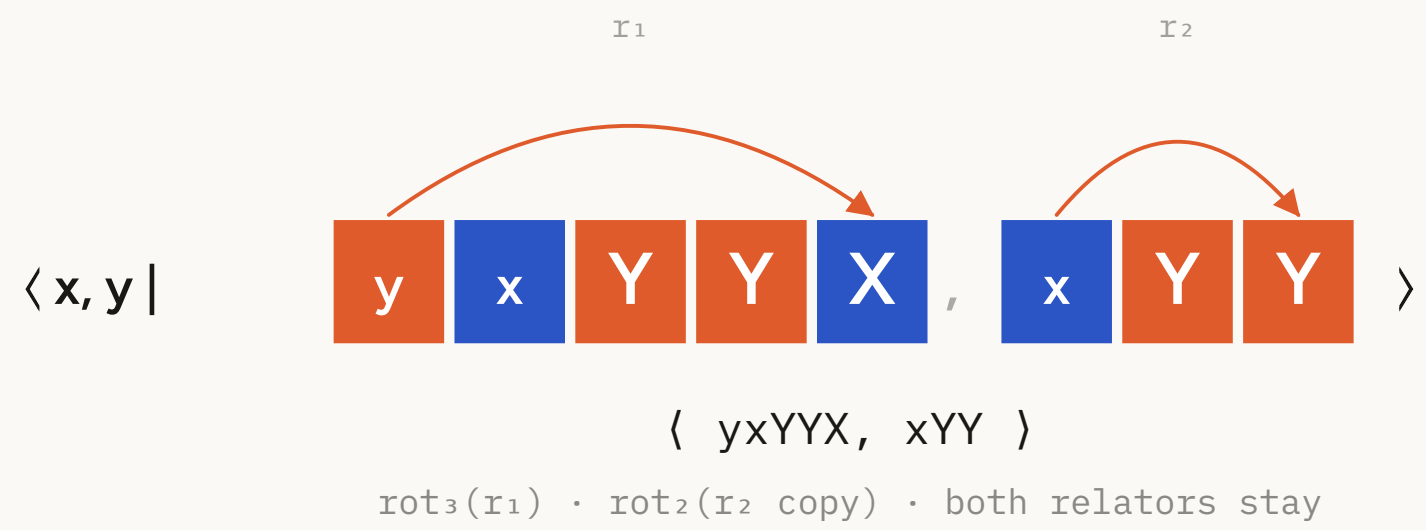
parts of one S-move

rotate r_1 · r_2 copy · concatenate · into r_1



One S-move

then to trivial



one S-move

$k_1 = 3 \cdot k_2 = 2$
into r_1

parts of one S-move

rotate r_1 · r_2 copy · concatenate · into r_1



One S-move

then to trivial

new r_1 = product

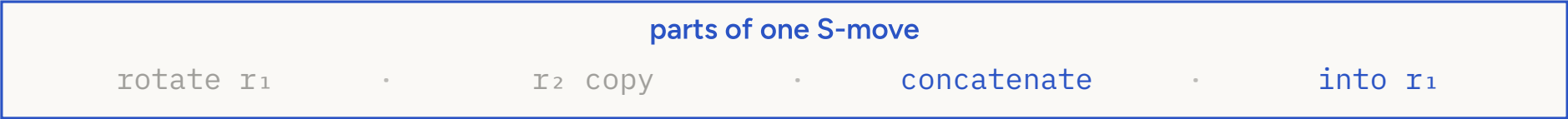
r_2 stays

one S-move

into r_1



$yxYYYX \cdot xYY$
 $r_2 \text{ copy} \cdot r_2 \text{ still } YYx$



One S-move

then to trivial

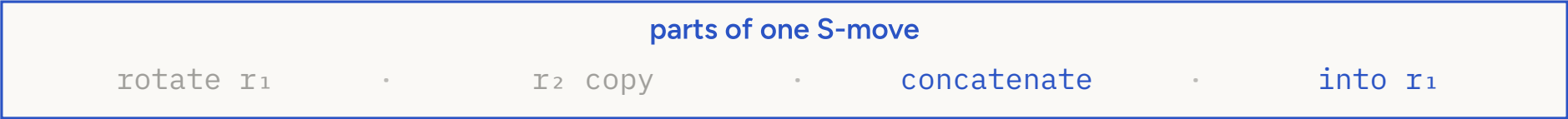
one S-move

Xx cancel



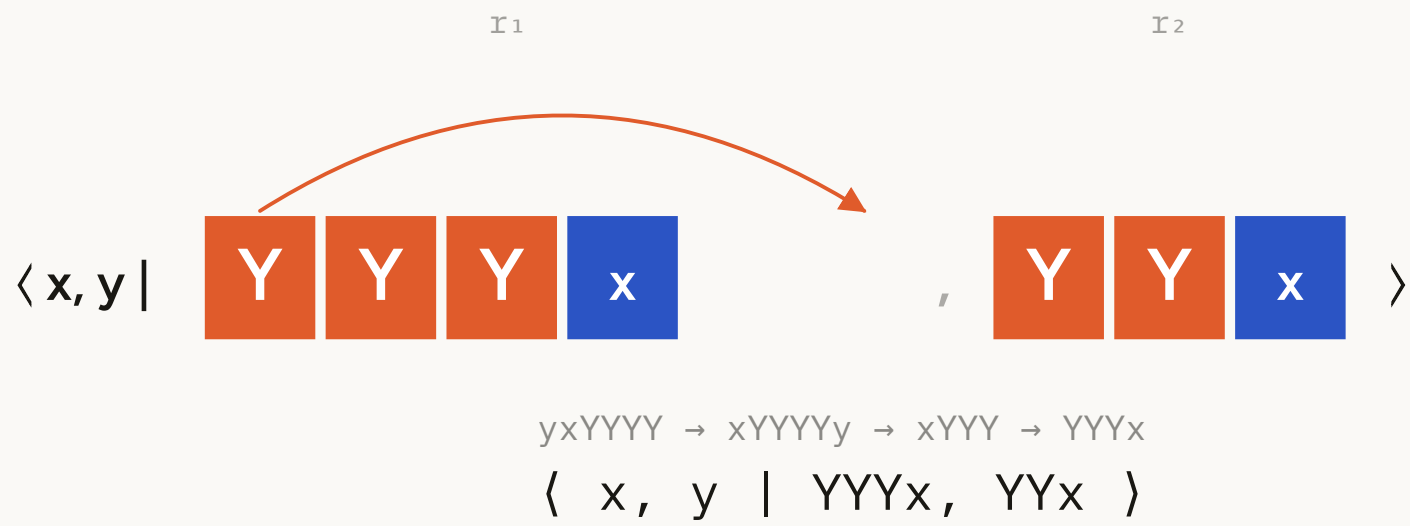
Xx cancel

$$r_1 = yxYYYY \cdot r_2 = YYx$$



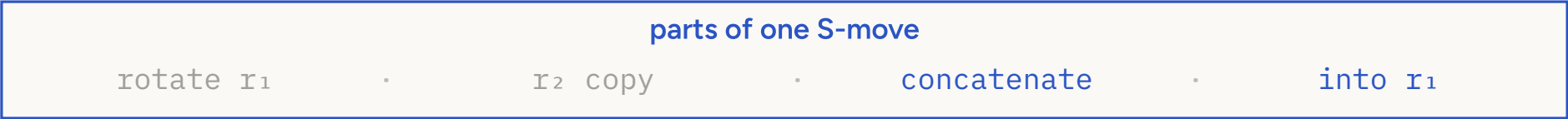
One S-move

then to trivial



one S-move

cyclic reduce · Booth



One S-move

then to trivial

S-move 2

invert $r_2 \cdot k_1 = 0 \cdot k_2 = 0$
into r_1

$\langle x, y |$



,



$\rangle$

$YYYx \cdot Xyy$
 $\langle Y, YYx \rangle$



One S-move

then to trivial

S-move 3

invert $r_2 \cdot k_1 = 0 \cdot k_2 = 0$
into r_2

$\langle x, y |$  ,   $\rangle$

$Y \cdot Xyy \rightarrow \text{cyclic reduce}$
 $\langle Y, Xy \rangle$



One S-move

then to trivial

$$\langle x, y \mid \text{Y}, \text{X} \rangle$$

$Y \cdot Xy \rightarrow \text{cyclic reduce}$

$$\langle Y, X \rangle \sim \langle x, y \mid x, y \rangle$$

trivial

S-move 4 · into r₂
k₁ = 0 · k₂ = 0



Four S-moves, fourteen elementary AC

the formula

$$\langle \text{YYXyx}, \text{YYx} \rangle \rightarrow \langle \text{YYYx}, \text{YYx} \rangle \rightarrow \langle \text{Y}, \text{YYx} \rangle \\ \langle \text{Y}, \text{Xy} \rangle \rightarrow \langle \text{Y}, \text{X} \rangle$$

14

elementary AC moves

=

4

S-moves

$$8 + 3 + 2 + 1 \cdot \text{cyclic reduce free}$$

Stable Andrews–Curtis

the main work this summer

stabilize

$\langle x, y \mid r_1, r_2 \rangle$



$\langle x, y, z \mid r_1, r_2, z \rangle$

destabilize

$\langle x, y, z \mid r_1, r_2, z \rangle$



$\langle x, y \mid r_1, r_2 \rangle$

two extra moves · with these, does a path always exist?



Stable Andrews–Curtis

the main work this summer

Lemma 11

Shehper et al.

$$\langle x_1, \dots, x_n, y \mid r_1, \dots, r_n, y^{-1}w \rangle$$

~

$$\langle x_1, \dots, x_n \mid r_i \text{ with } y \text{ replaced by } w \rangle$$

if an extra letter is defined as a word, substitute and drop it



Stable Andrews–Curtis

the main work this summer

Lemma 11

Shehper et al.

$$\langle x, y, z \mid z \cdot z \cdot x, y \cdot z \cdot Y, Z \cdot Xy \rangle$$

the stable form $\cdot z \equiv Xy$



Stable Andrews–Curtis

the main work this summer

Lemma 11

Shehper et al.

$$\langle x, y, z \mid z \cdot z \cdot x, y \cdot z \cdot Y, Z \cdot Xy \rangle$$



replace every z by Xy

$$\langle x, y, z \mid Xy \cdot Xy \cdot x, y \cdot Xy \cdot Y, Yx \cdot Xy \rangle$$

$$Yx \cdot Xy = 1$$



Stable Andrews–Curtis

the main work this summer

Lemma 11

Shehper et al.

$\langle x, y, z \mid Xy \cdot Xy \cdot x, y \cdot Xy \cdot Y, Yx \cdot Xy \rangle$



destabilize · drop z

$\langle x, y \mid Xy \cdot Xy \cdot x, y \cdot Xy \cdot Y \rangle$

two generators again



Change of variables

$$z = Xy$$

$$z = w(x, y)$$

$$\langle x, y \mid YYXy, Yx \rangle$$

$$z = \begin{bmatrix} X & y \end{bmatrix}$$

$$z^{-1} = \begin{bmatrix} Y & x \end{bmatrix}$$



Change of variables

$z = Xy$

stabilize

$\langle x, y \mid YYXyx, Yx \rangle$



$\langle x, y, z \mid YYXyx, Yx, ZXy \rangle$

$z = Xy$



Change of variables

$$z = Xy$$

substitute · isolate x

$$\langle x, y, z \mid YYXyx, Yx, ZXy \rangle$$

$$z \rightarrow Xy$$

$$Z \rightarrow Yx$$



$$\langle x, y, z \mid YYzx, Z, ZXy \rangle$$

$$YYzx = 1 \rightarrow x = Zyy$$



Change of variables

$$z = Xy$$

substitute x · then destabilize

$$\langle x, y, z \mid YYz\textcolor{brown}{x}, Z, Z\textcolor{brown}{X}y \rangle$$

$$\textcolor{brown}{x} = Zyy \cdot \textcolor{brown}{X} = YYz$$



$$\langle x, y, z \mid YYz\textcolor{brown}{Z}yy, Z, ZYY\textcolor{brown}{z}y \rangle$$

$$YYz\textcolor{brown}{Z}yy \rightarrow 1$$



$$\langle y, z \mid Z, ZYY\textcolor{blue}{z}y \rangle$$

destabilize · drop x



Change of variables

$$z = Xy$$

relabel

$$\langle y, z \mid Z, ZYyzy \rangle$$



$$\langle x, y \mid Y, YXXyx \rangle$$

$$y, z \rightarrow x, y$$



A benchmark subset of 1,190 presentations

640 solved · 550 unsolved

1,190

Miller-Schupp presentations



640

solved

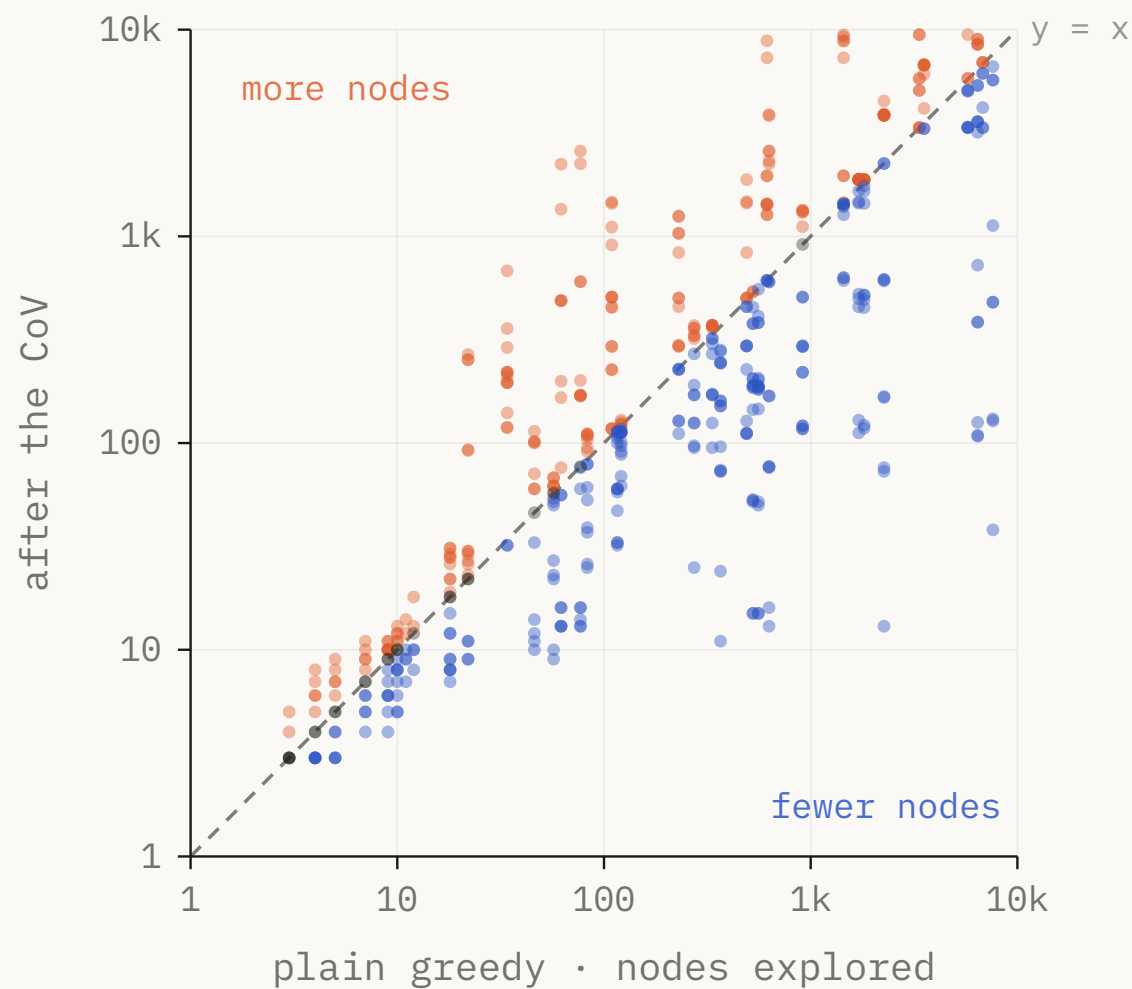
550

unsolved

previous paper · 261 classes

Cost of a change of variables

60 presentations · 45 CoVs ·
budget 10,000



each dot is one CoV

546 CoVs that solved

288 fewer nodes

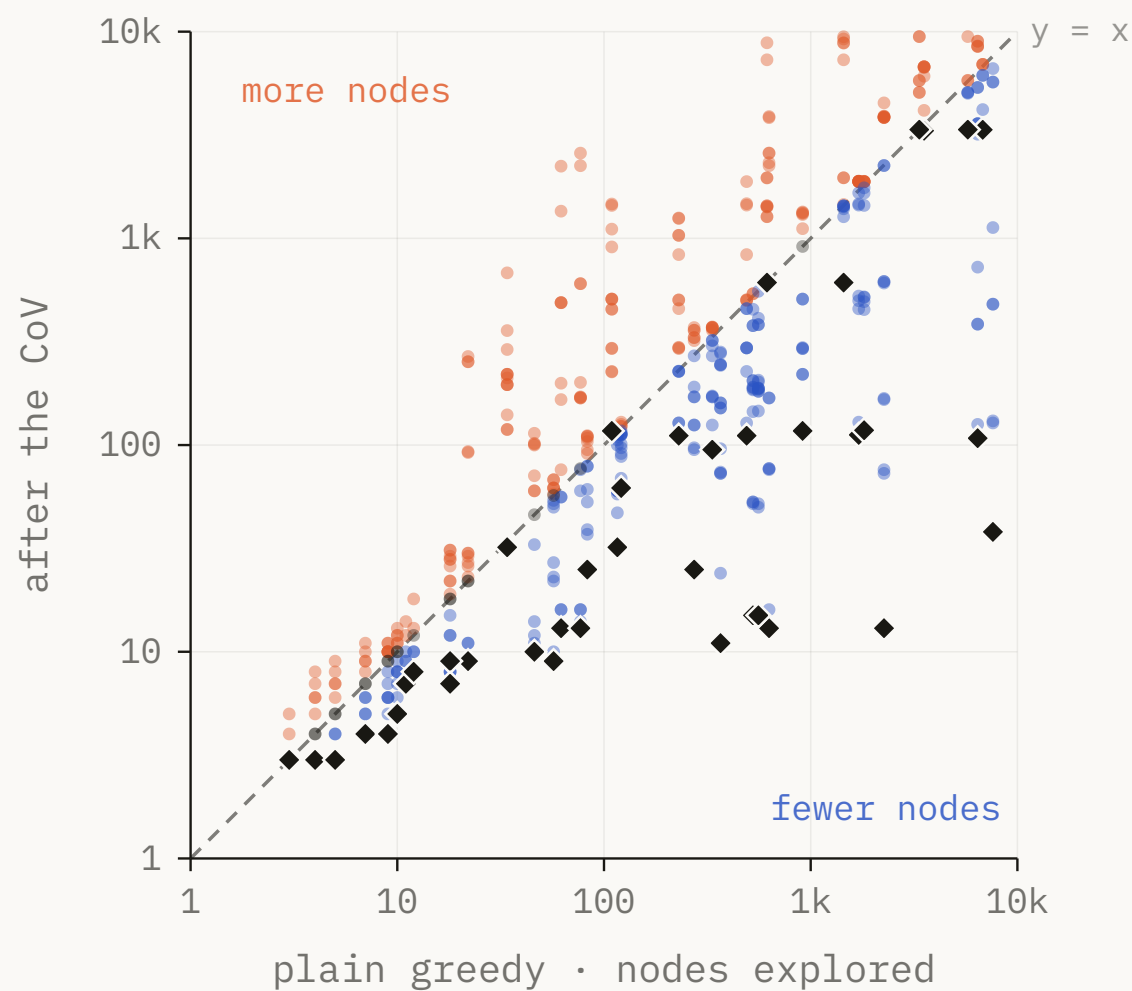
235 more nodes

23 no change



Cost of a change of variables

60 presentations · 45 CoVs ·
budget 10,000



each dot is one CoV

546 CoVs that solved

288 fewer nodes

235 more nodes

23 no change

◆ best CoV per presentation

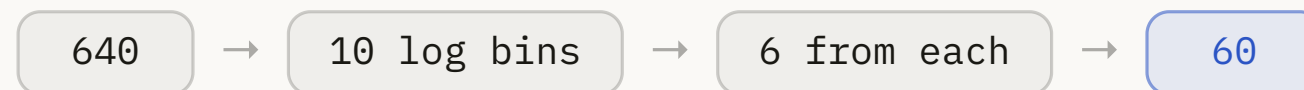
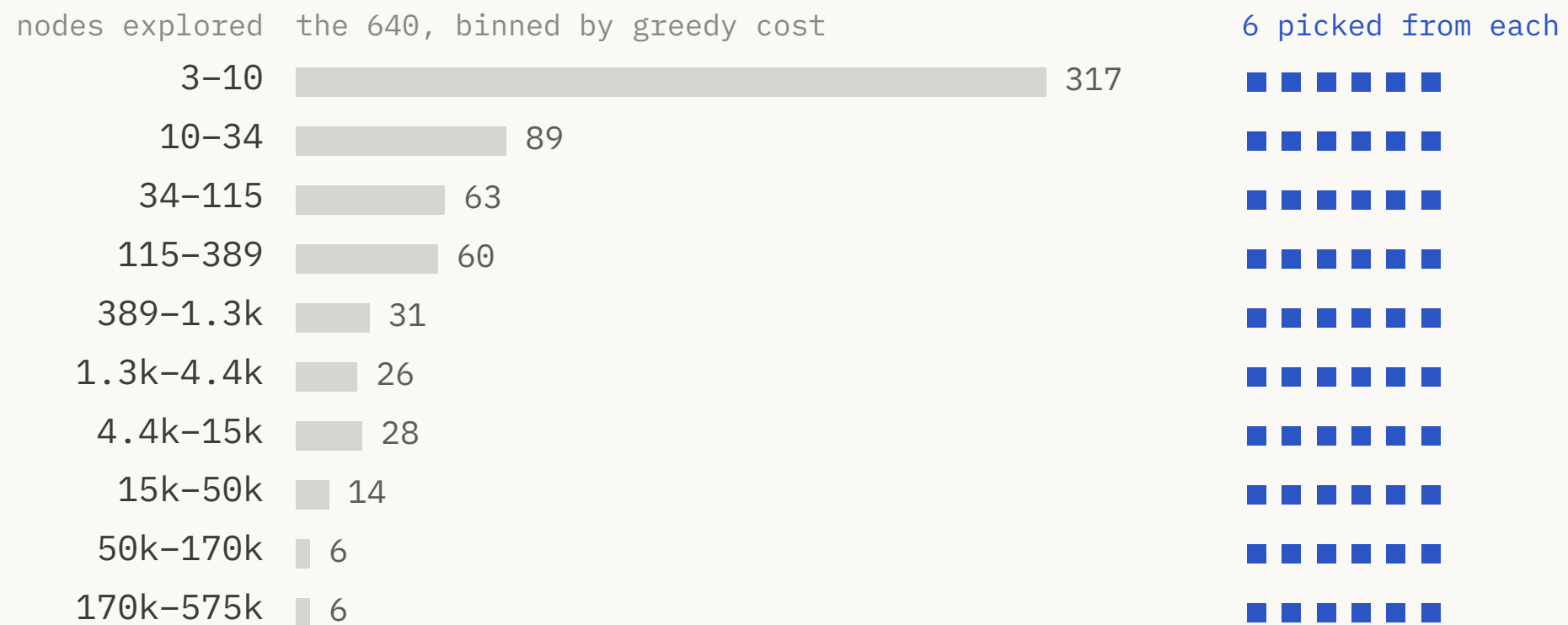
beats greedy on 37 of 40
on 3, no CoV beat greedy

so the choice matters



The subset-60 benchmark

640 → 60 · then greedy vs
best CoV



bins are $\times 3.37$ wide · the 60 cover 45 distinct Aut classes



The subset-60 benchmark

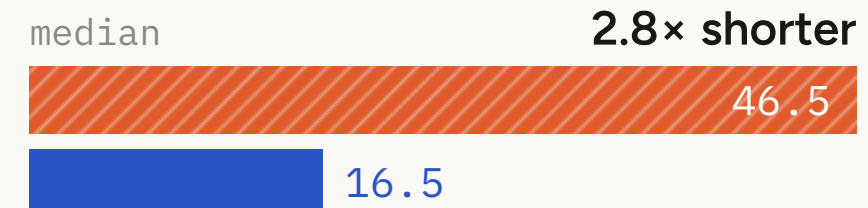
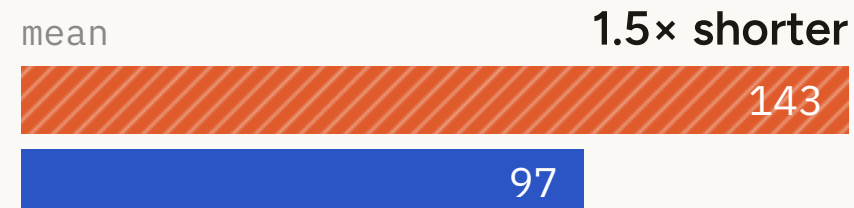
640 → 60 · then greedy vs
best CoV

greedy vs best CoV · all 60 solved by both

nodes explored



path length



plain greedy  best CoV 

each panel scaled to its own greedy bar

greedy budget 1,000,000 · best CoV = cheapest of 11-173 starts, budget 20,000 · CoV path counted in the new coordinates



261 problems are really 124

261 → 124

unsolved

< Y Y Y Y X y y y x , Y X Y x x >

≠

unsolved

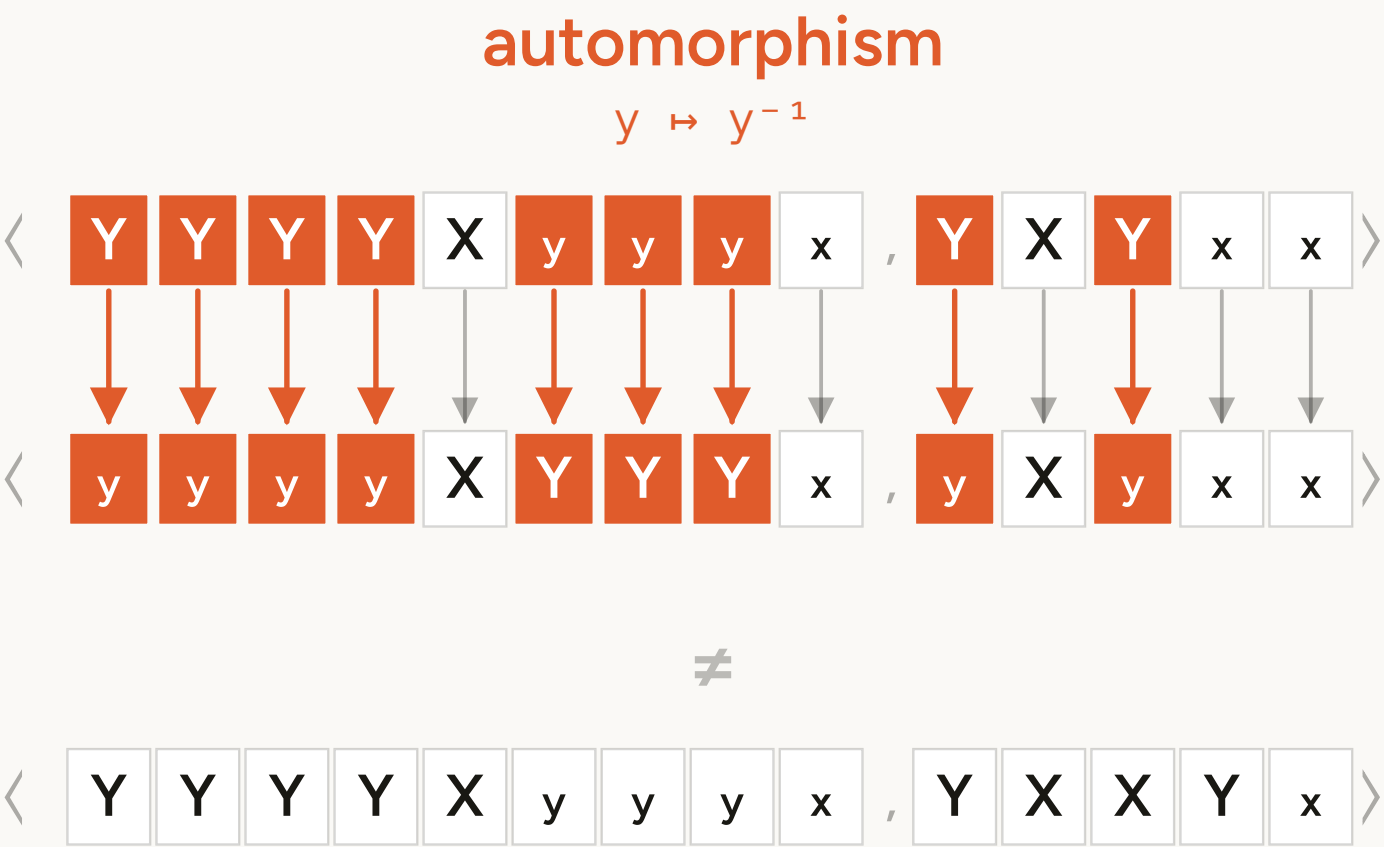
< Y Y Y Y X y y y x , Y X X Y x >

261 → 124



261 problems are really 124

261 → 124



261 → 124



261 problems are really 124

261 → 124

invert

each relator → reverse, invert letters

$\langle x, y |$

X	y	y	y	x	Y	Y	Y	Y
---	---	---	---	---	---	---	---	---

 ,

X	X	Y	x	Y
---	---	---	---	---

 $\rangle$

≠

$\langle x, y |$

Y	Y	Y	Y	X	y	y	y	x
---	---	---	---	---	---	---	---	---

 ,

Y	X	X	Y	x
---	---	---	---	---

 $\rangle$

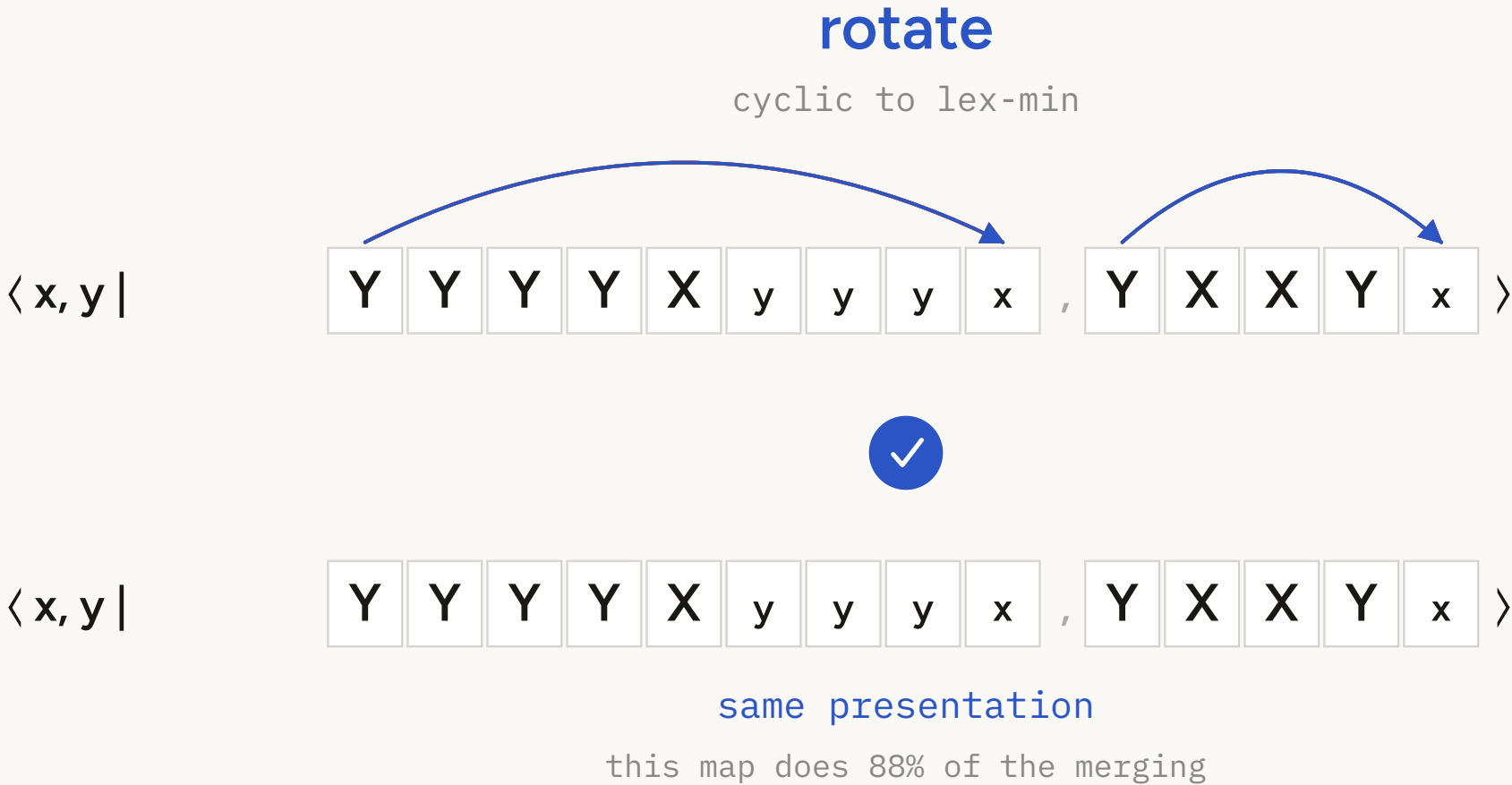
$\langle XyyyxYYYY, XXYxY \rangle$ · other already canonical

261 → 124



261 problems are really 124

261 → 124

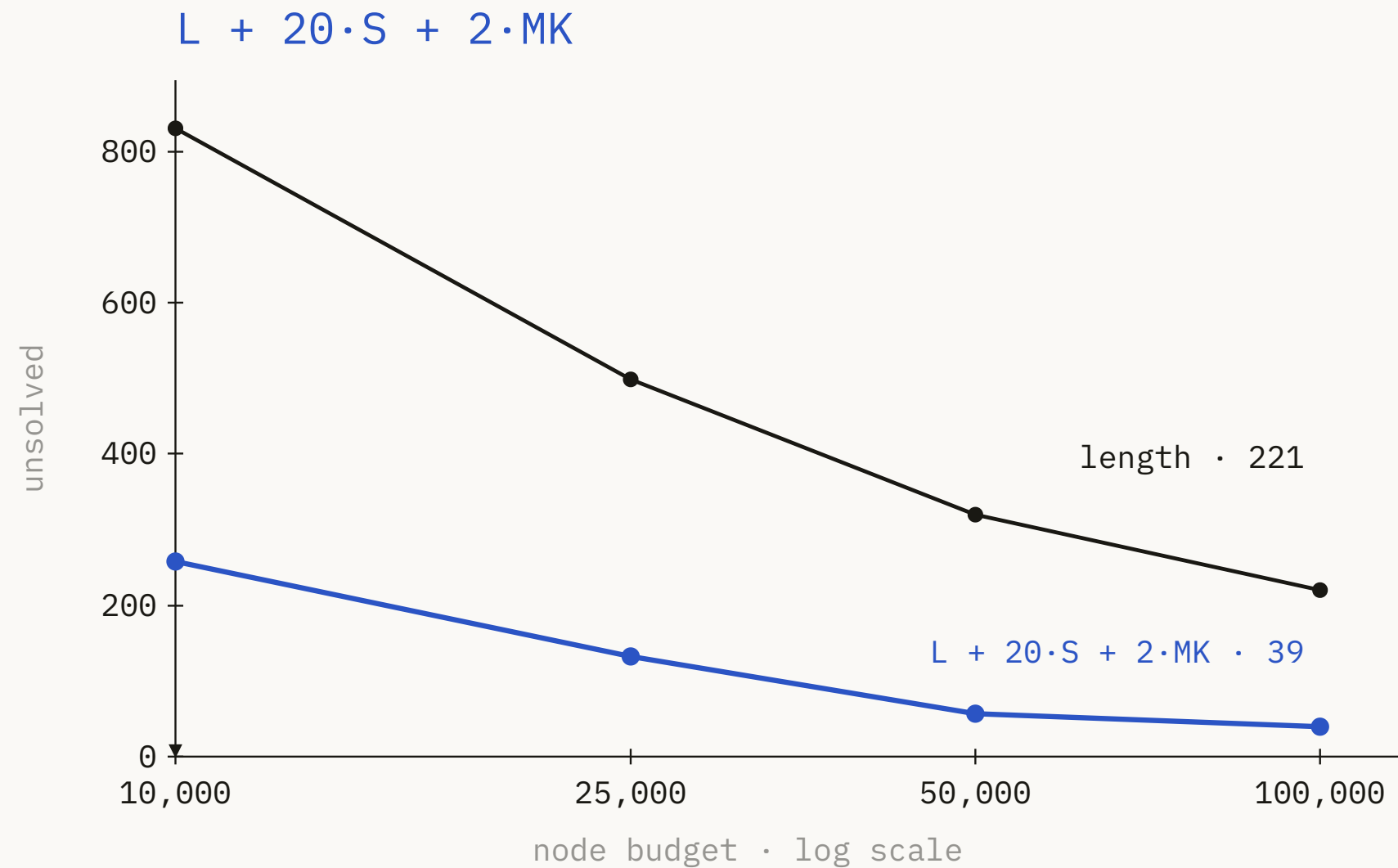


261 → 124



A better heuristic: 221 → 39

70,723 Aut-minimal orbits



95 → 277 / 316
hardest bin solved

563.3 → 263.2
mean nodes

23.5 → 26.6
mean path length
3 moves longer

Abelian mass heuristic

all 640 Miller-Schupp · budget
100,000

$$\text{mass} = |\Sigma x \ r_1| + |\Sigma y \ r_1| + |\Sigma x \ r_2| + |\Sigma y \ r_2|$$

one pass · no search

example

$\langle x, y \mid YYXyx, Yx \rangle$

$r_1 \cdot \Sigma x = 0, \Sigma y = -1$

$r_2 \cdot \Sigma x = +1, \Sigma y = -1$

$\text{mass} = 0 + 1 + 1 + 1 = 3$

trivial pair = 2



Abelian mass heuristic

all 640 Miller-Schupp · budget
100,000

$$\text{mass} = |\Sigma x \ r_1| + |\Sigma y \ r_1| + |\Sigma x \ r_2| + |\Sigma y \ r_2|$$

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$r_2 \cdot \Sigma x = +1, \Sigma y = -1$

$$\text{mass} = 0 + 1 + 1 + 1 = 3$$

trivial pair = 2

mass $\rightarrow$ length $\rightarrow$ S

tied at rank 1: 51 $\rightarrow$ 3 $\rightarrow$ 1



Abelian mass heuristic

all 640 Miller-Schupp · budget
100,000

$$\text{mass} = |\Sigma x \ r_1| + |\Sigma y \ r_1| + |\Sigma x \ r_2| + |\Sigma y \ r_2|$$

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example

$\langle x, y \mid YYXyx, Yx \rangle$

$r_1 \cdot \Sigma x = 0, \Sigma y = -1$

$r_2 \cdot \Sigma x = +1, \Sigma y = -1$

$$\text{mass} = 0 + 1 + 1 + 1 = 3$$

trivial pair = 2

mass $\rightarrow$ length $\rightarrow$ S

tied at rank 1: 51 $\rightarrow$ 3 $\rightarrow$ 1

plain greedy · 3,176,297 nodes · 634/640 at 100k



(640 needs 1M)

mass-ranked CoV, rank 1 · 420,426 nodes · 640/640



7.6×

fewer nodes



Two letters away from solved

AK(3) · Akbulut-Kirby 1985

$\langle x, y |$

Y	X	y	x	Y	x
---	---	---	---	---	---

 ,

Y	Y	Y	Y	x	x	x
---	---	---	---	---	---	---

 $\rangle$



solved · 12 moves

$\langle x, y |$

Y	X	Y	x	y	x
---	---	---	---	---	---

 ,

Y	Y	Y	Y	x	x	x
---	---	---	---	---	---	---

 $\rangle$



never solved

Still unsolved

124 classes · AK(3)

CoV sweep @ 50k · 0 / 124 solved

iterative CoV · 39 / 124 reduced
36 of the 39 from the μ -ladder

shorter start · same floor · 0 solved

567M nodes · 182 hours · 0 of 124 solved

!

AK(3)

floor · total length 13

stably trivial · 12

Two ideas solved all 640

heuristic – $L + 20 \cdot S + 2 \cdot MK$

start – lowest abelian mass



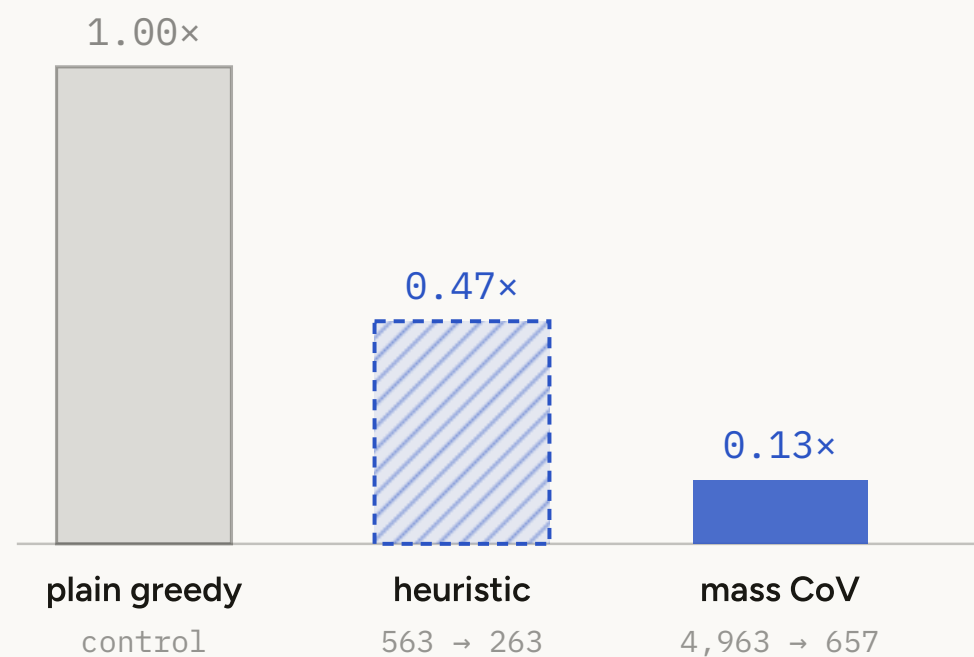
640 / 640

solvable set, 7.6× cheaper

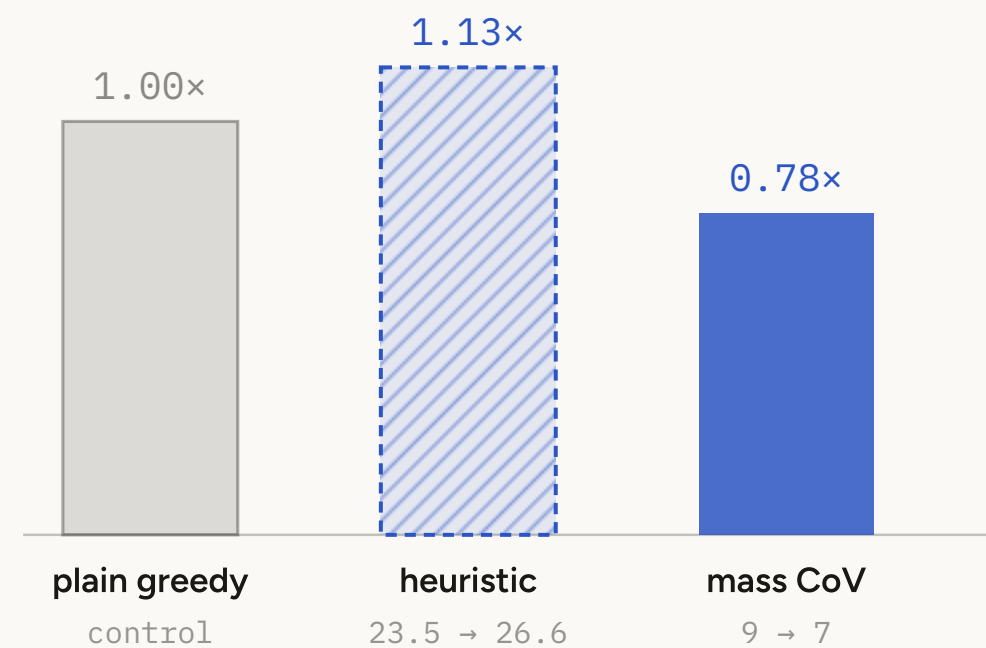


124 open problems

nodes explored



path length



each idea against its own greedy control · heuristic on 70,723 Aut orbits · CoV on all 640

What comes next

two directions

- 1 **PPO with heuristic**
- 2 **Improved Benchmark besides MS1190**

Thank you

Prof. Sergei Gukov

Math AI Group · Caltech

Lucas Fagan

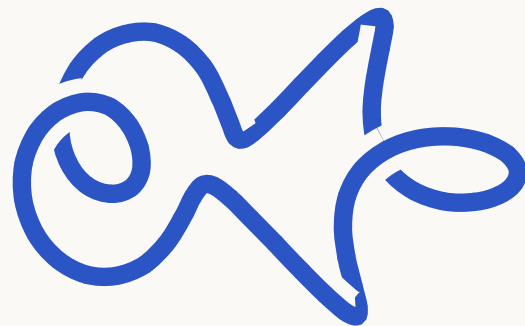
Giorgi Butbaia

Michele Tarquini

mentors · Math AI Group · Caltech

Claude

Anthropic



Questions

Feedback

Summer Seminar Day 2026



Avigya Paudel

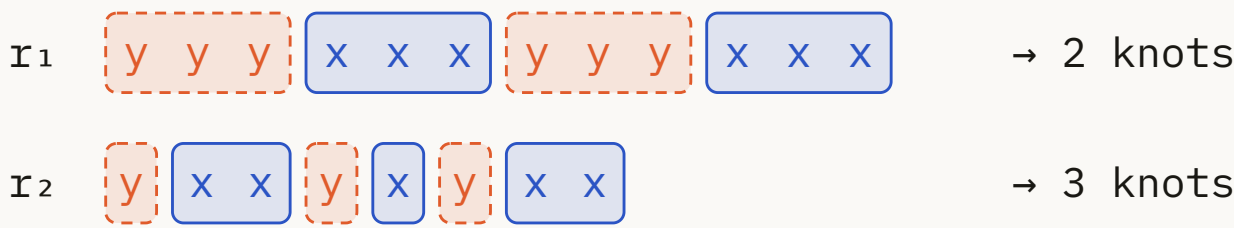
scan to send feedback

Appendix · S and max knots

$$L + 20 \cdot S + 2 \cdot MK$$

worked example

knots = blocks



MK = max(2, 3) = 3

S = smaller mean block

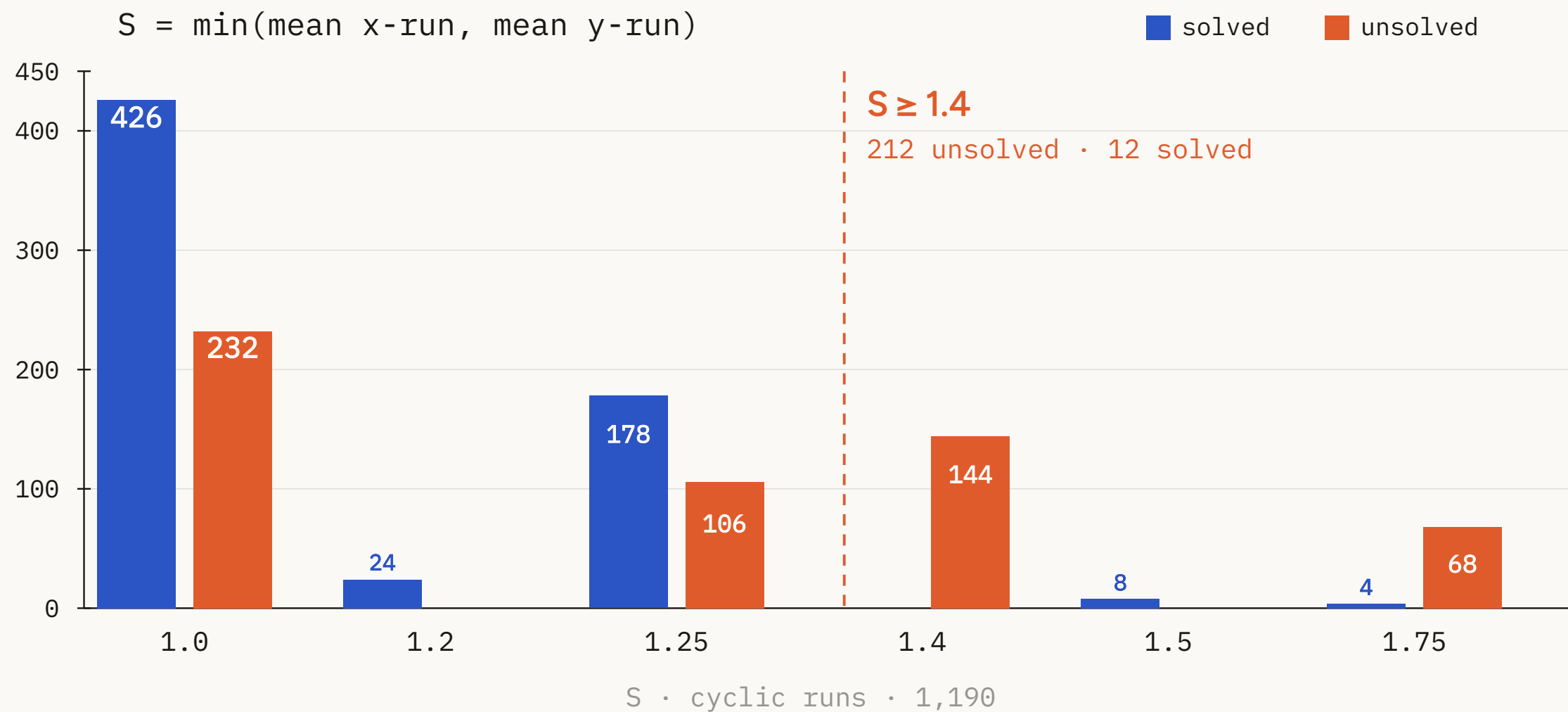
	r ₁		r ₂			
x-blocks	3	3	2	1	2	mean 2.2
y-blocks	3	3	1	1	1	mean 1.8

S = min(2.2, 1.8) = 1.8



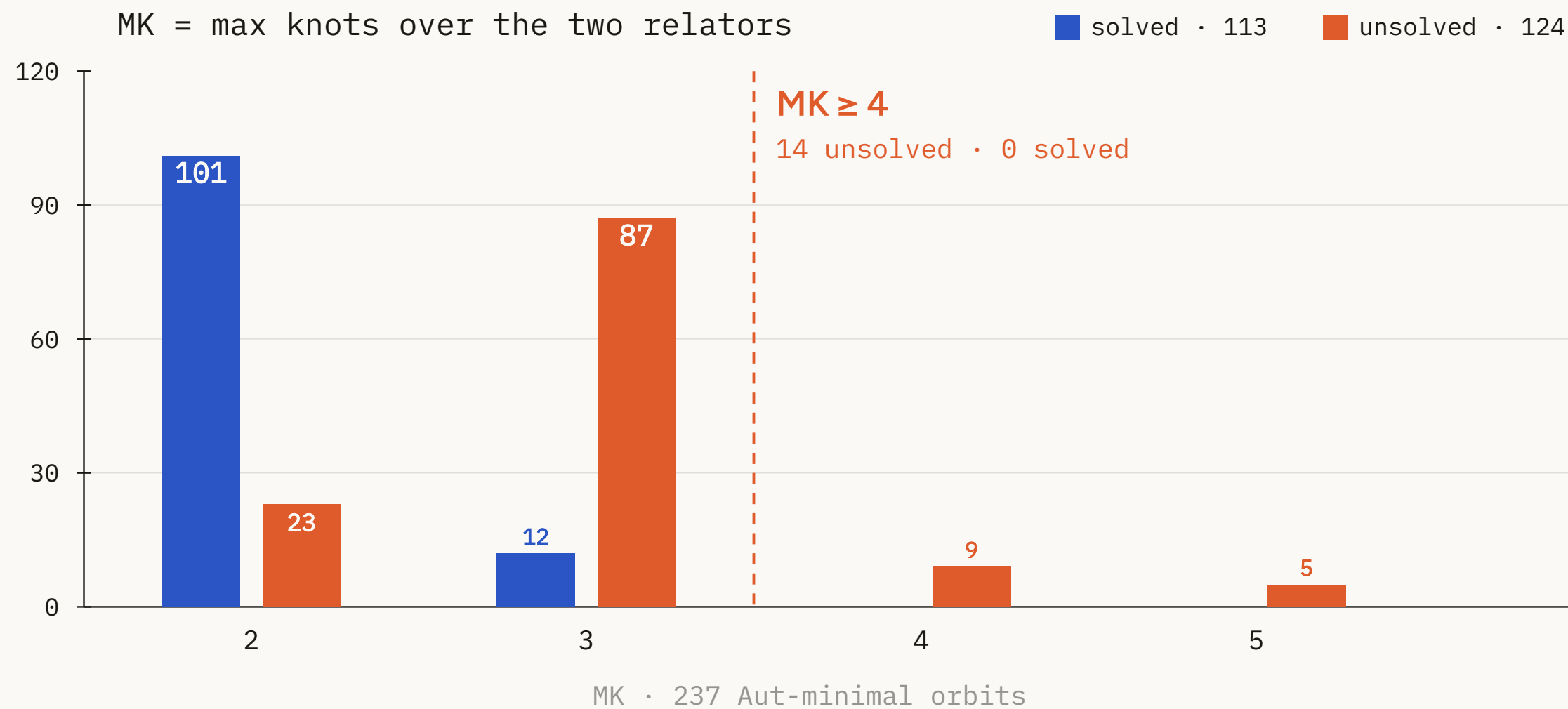
Appendix · S and max knots

$$L + 20 \cdot S + 2 \cdot MK$$



Appendix · S and max knots

$$L + 20 \cdot S + 2 \cdot MK$$



Appendix · cheaper nodes, longer paths

length · 23.5

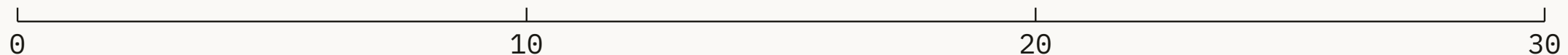


23.5

heuristic · 26.6



26.6



the heuristic spends moves to save nodes · 70,502 both-solved

Appendix · raising the cap doesn't help

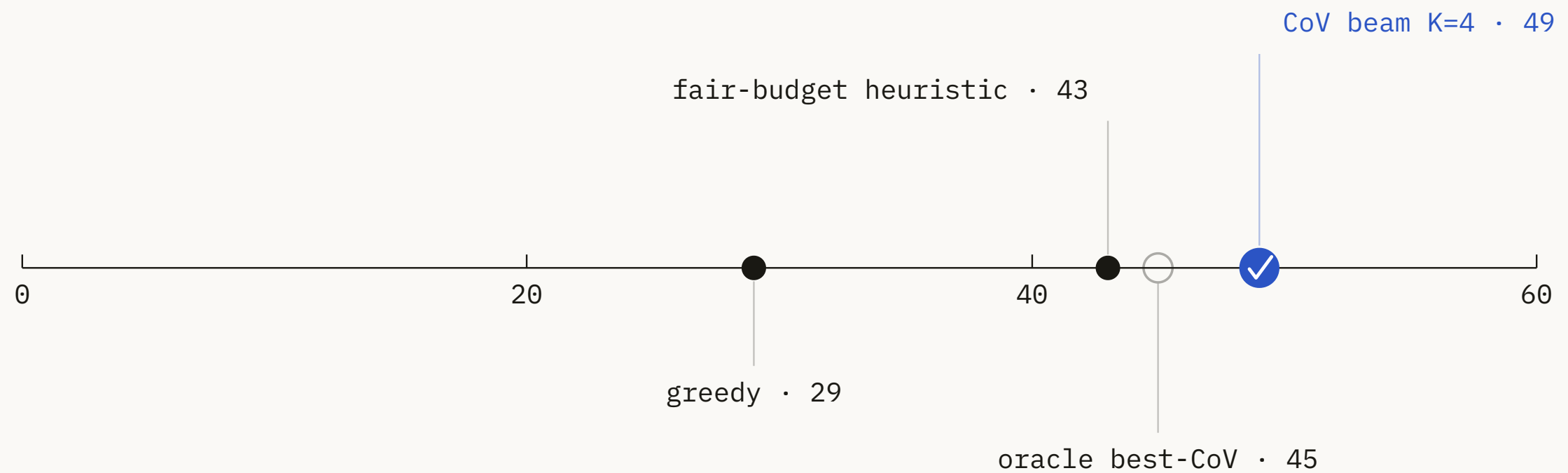
L = 24 ≡ L = 48

byte-identical visited sets · 16 / 16 runs

0 / 1,800

over-cap candidates solve

Appendix · a CoV beam beats the oracle

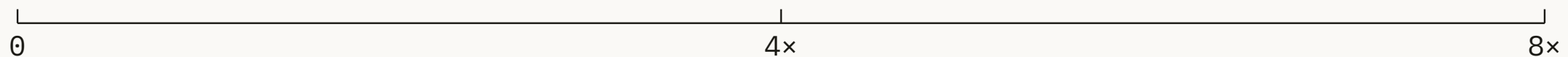


Appendix · relabels inflated the CoV headline

all CoVs · median cut 7.58x



non-automorphic only · 3.62x



87% of CoV candidates are relabelings

Appendix · AK(3) in the literature



Appendix · how we know the group is 1

1

balanced

2 generators, 2 rules

2

abelianization

letter-counts · det ± 1

3

helper letter

add $z \rightarrow$ it trivializes

the group is 1

AC forbids the helper letter

AK(3) is stably trivial · still no AC path

Appendix · what abelian mass does not do

60 presentations · ~10⁹ CoVs
each · budget 10,000

1 rank one presentation's own candidates

AUC 0.922 picks solvers from non-solvers · positive on 52 / 52

q +0.81

2 screen which presentation to attack

pooled over 6,722 rows · length reads +0.46 here and is the wrong key – Simpson's paradox

r +0.11

3 pick the one z to run

a median of 6 candidates tie at the minimum · best-of-3 reaches 52 / 60, the oracle set

47.4 / 60

it ranks inside a presentation – it does not screen between them, and it shortlists rather than picks